

Digital Asset Management

Functional Requirements and Architecture Specification

Technical Specification

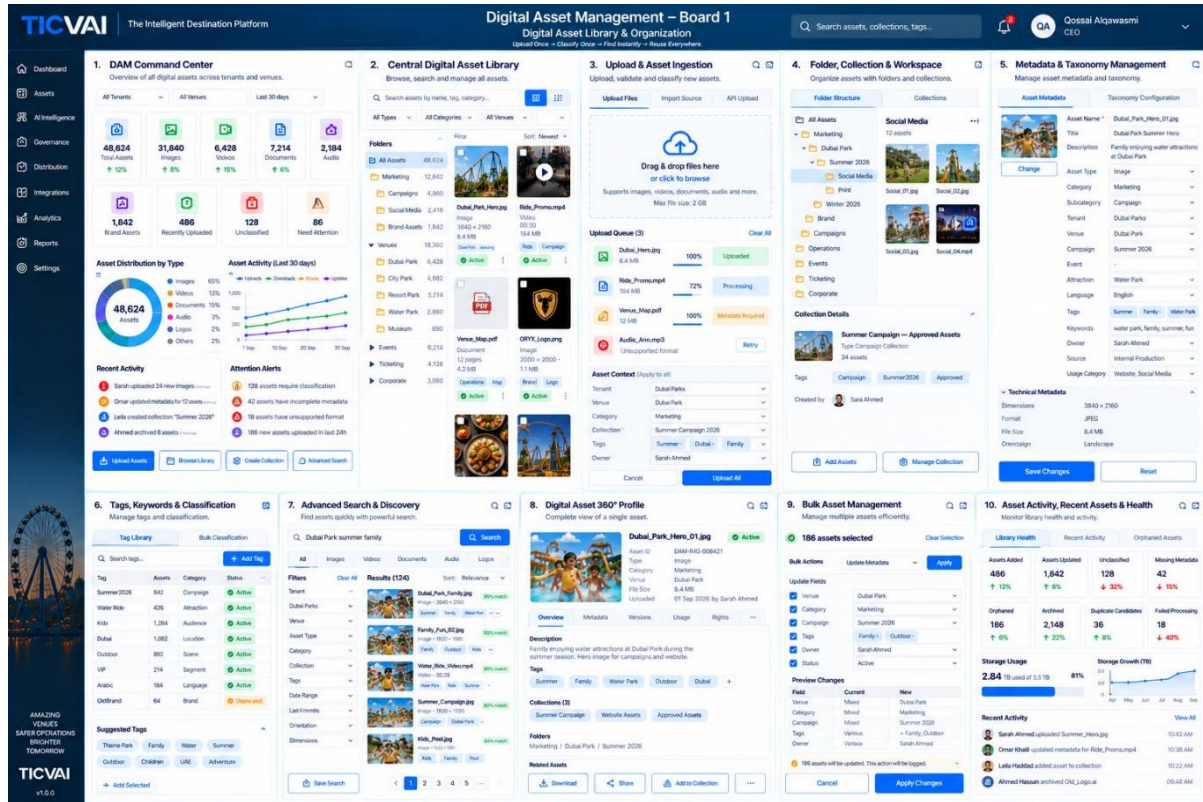
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Board 1 — Digital Asset Library & Organization

Board Objective

Create the **central digital media repository for TICVAI**, allowing authorized users to upload, organize, classify, search, preview, manage, and reuse digital assets across tenants, venues, events, attractions, ticketing, marketing, CRM, websites, mobile applications, POS, kiosks, and other TICVAI modules.



Board 1 should establish the **master digital asset record and organization layer**. AI intelligence/versioning will be handled in Board 2, rights/governance in Board 3, and distribution/integration in Board 4.

The core principle should be:

Upload once → classify once → find instantly → reuse everywhere.

Screen 1 — Digital Asset Management Command Center

Purpose

Provide a centralized operational dashboard showing the complete digital asset estate across the authorized tenant and venue scope.

Global Filters

Support:

Tenant | Venue | Asset Type | Category | Status | Owner | Date Range

KPI Cards

Display:

- Total Digital Assets
- Images
- Videos

- Documents
- Audio Files
- Brand Assets
- Recently Uploaded
- Storage Used
- Unclassified Assets
- Assets Requiring Attention

Example:

KPI	Value
Total Assets	48,624
Images	31,840
Videos	6,428
Documents	7,214
Brand Assets	1,842
Recently Uploaded	486
Unclassified	128
Storage Used	2.84 TB

Asset Activity

Show:

- Uploads
- Updates
- Downloads
- Shares
- Recently Used

Asset Distribution

Charts by:

- File type
- category
- venue
- department
- source
- status

Attention Panel

Examples:

- ⚠ 128 uploaded assets require classification.
- 🕒 42 assets have incomplete mandatory metadata.
- 🕒 186 new assets were uploaded during the last 24 hours.

Quick Actions

- Upload Assets
- Browse Library
- Create Collection

- **Bulk Manage**
- **Advanced Search**

Screen 2 — Central Digital Asset Library

Purpose

Provide the main workspace for browsing all digital assets available to the authorized user.

Library Views

Support:

- **Grid**
- **List**
- **Compact**
- **Gallery**

Asset Cards

Each card should show:

- Thumbnail
- asset name
- asset type
- category
- dimensions/duration where applicable
- file size
- status
- owner
- updated date

Example:

Dubai_Park_Hero_01.jpg

Image

3840 × 2160

8.4 MB

Category:

Marketing / Dubai Park

Status:

 **Active**

Asset Types

Support at minimum:

- **Images**
- **Videos**
- **Documents**
- **Audio**
- **Logos**
- **Brand Files**
- **Marketing Creative**
- **Ticket Media**

Venue Media

and extensible future types.

Filters

Allow filtering by:

- Tenant
- Venue
- Asset Type
- Category
- Collection
- Tags
- Owner
- Created Date
- Updated Date
- Status
- File Format

Sort

Support:

- Newest
- Oldest
- Recently Updated
- Name
- File Size
- Most Used

Actions

- Preview
- View Details
- Download
- Add to Collection
- Move

Archive

subject to permissions.

Screen 3 — Upload & Asset Ingestion Workspace

Purpose

Provide a controlled process for adding digital assets into TICVAI.

Upload Methods

Support:

- Drag & Drop
- Browse Files
- Bulk Upload
- Import from Approved Source

API Upload

where configured.

Upload Queue

Example:

File	Type	Size	Progress	Status
Dubai_Hero.jpg	Image	8.4 MB	100%	Uploaded
Ride_Promo.mp4	Video	184 MB	72%	Processing
Venue_Map.pdf	Document	12 MB	100%	Metadata Required

Upload Validation

Validate:

- Supported file format
- maximum file size
- file integrity
- mandatory metadata
- duplicate indicators
- security/malware scanning status where applicable

Upload Context

User should be able to define:

- Tenant
- Venue
- Category
- Collection
- Tags
- Owner

Purpose

before or after upload.

Bulk Metadata

Example:

Apply Dubai Park / Summer Campaign 2026 to all 24 uploaded files.

Processing Status

Use:

Uploading



Processing



Validation



Metadata



Ready

Failed Upload

Clearly show:

Reason

and:

Retry

Important

Board 1 may **detect or flag a possible duplicate during ingestion**, but advanced duplicate intelligence and similarity analysis belong to Board 2.

Screen 4 — Folder, Collection & Workspace Management

Purpose

Allow users to organize assets without relying only on physical file-storage concepts.

TICVAI should support both:

Folder Structure

and:

Collections

This distinction is important.

Folder

Represents organizational hierarchy.

Example:

Marketing

↓

Dubai Park

↓

Summer 2026

↓

Social Media

Collection

Represents a logical group of assets that may exist in different folders.

Example:

Summer Campaign — Approved Assets

could contain:

- Logo
- hero image
- Instagram creative
- video
- venue map

without physically duplicating those files.

Collection Types

Support:

- **Manual Collection**
- **Dynamic Collection**
- **Campaign Collection**
- **Brand Collection**
- **Event Collection**

Dynamic Collection Example

Automatically include:

Asset Type = Image

Venue = Dubai Park

Tag = Summer2026

Status = Active

Actions

- Create Folder
- Create Collection
- Move Assets
- Add Assets
- Remove from Collection
- Rename
- Archive

Critical Rule

Adding an asset to multiple collections must **not create multiple copies of the original file**.

Screen 5 — Metadata & Taxonomy Management

Purpose

Create the structured metadata model that makes TICVAI's DAM searchable and scalable.

Asset Metadata

Support:

- Asset Name
- Title
- Description
- Asset Type
- Category
- Subcategory
- Tenant
- Venue
- Department
- Campaign
- Event
- Attraction
- Language
- Tags
- Keywords
- Owner
- Source
- Creation Date
- Upload Date

Technical Metadata

Automatically capture where possible:

For images:

- Dimensions
- resolution
- format
- file size
- orientation

For videos:

- Duration
- resolution
- codec
- frame rate
- file size

For audio:

- Duration
- format
- bitrate

For documents:

- Format
- pages where detectable
- file size

Taxonomy Management

Administrators should create controlled taxonomies.

Example:**Destination**

- Dubai Park
- Water Park
- Museum

Asset Purpose

- Marketing
- Ticket Media
 - Website
- Mobile App
 - POS
 - Kiosk
 - Internal

Mandatory Metadata

Different asset categories may require different fields.

Example:**Marketing Image****Requires:**

Venue

Campaign

Language

Usage Category

while:

Venue Map

may require:

Venue

Version

Effective Date

Configuration

Metadata fields must be configurable rather than hardcoded.

Screen 6 — Tags, Keywords & Classification

Purpose

Provide flexible classification in addition to formal taxonomy.

Tag Management

Users with permission can:

Add Tag

Remove Tag

Bulk Tag

Merge Tag

Rename Tag

Example Tags

Family

Water Ride

Summer

Dubai

Outdoor

Kids

VIP

Arabic

Tag Suggestions

Board 1 should support manual and configured suggestions.

Advanced:

AI Auto-Tagging

belongs primarily to Board 2.

Tag Library

Display:

Tag	Assets	Category	Status
Summer2026	842	Campaign	Active
Water Ride	426	Attraction	Active

Tag	Assets	Category	Status
Kids	1,284	Audience	Active
OldBrand	64	Brand	Deprecated

Tag Governance

Prevent uncontrolled variations such as:

Waterpark

Water Park

water_park

if the administrator wants a controlled vocabulary.

TICVAI can recommend:

- Use existing tag **Water Park**?

Bulk Classification

Select 100 assets and apply:

Venue: Dubai Park

Category: Marketing

Campaign: Summer 2026

Tags: Family, Outdoor

Screen 7 — Advanced Search & Discovery

Purpose

Allow users to find assets quickly even when the library contains hundreds of thousands or millions of records.

Global Search

Search across:

- Asset name
- title
- description
- metadata
- tags
- keywords
- filename
- category
- collection
- owner

Search Example

Dubai Park summer family

Search Results

Display matching assets with:

- Thumbnail
- title
- type

- venue
- tags
- updated date
- relevance

Advanced Filters

Support:

- **Asset Type**
- **File Format**
- **Venue**
- **Category**
- **Collection**
- **Tag**
- **Owner**
- **Date**
- **Dimensions**
- **Orientation**
- **Duration**
- **Status**

Search Operators

Where appropriate:

AND

OR

Exclude

Example:

Dubai Park AND Summer NOT Archived

Saved Search

Allow:

- **Dubai Park Active Marketing Images**
- to be saved.

Recent Search

Display recent user searches.

Important Boundary

Board 1 provides structured search.

Board 2 will extend this into:

Semantic Search

Visual Similarity Search

Natural-Language Asset Discovery

Screen 8 — Digital Asset 360° Profile

Purpose

Provide one authoritative record for every digital asset.

Header

Example:

Dubai_Park_Hero_01.jpg

Asset ID: DAM-IMG-008421

Status:

 **Active**

Preview

Display a large preview appropriate to the asset type.

Core Information

Show:

- Asset ID
- title
- filename
- type
- format
- category
- owner
- tenant
- venue
- upload date
- last modified
- file size

Technical Information

Example:

3840 × 2160

JPEG

8.4 MB

Metadata

Display:

- **Campaign**
- **Event**
- **Attraction**
- **Language**
- **Tags**
- **Keywords**

Related Context

Show where relevant:

Collections

Folders

Related TICVAI Records

Tabs

I recommend:

Overview

Metadata**Versions****Usage****Rights****Sharing****Related Assets****Activity****Board Boundaries**

The profile can expose these tabs, but detailed functionality is provided by:

Versions → Board 2**Rights → Board 3****Sharing → Board 3****Distribution/Usage → Board 4**

This gives users one Asset 360° experience without duplicating engines.

Screen 9 — Bulk Asset Management Workspace**Purpose**

Allow enterprise users to efficiently manage large groups of digital assets.

Selection**Support:**

- **Select Individual**
- **Select Page**
- **Select Search Results**
- **Select Collection**

Bulk Actions**Allow, subject to RBAC:****Add Tags****Remove Tags****Change Category****Change Venue****Change Owner****Add to Collection****Move Folder****Update Metadata****Download****Archive****Example****186 Assets Selected****Apply:****Venue**

Dubai Park

Campaign

Summer 2026

Tag

Family

Preview Changes

Before applying:

Field	Current	New
Venue	Mixed	Dubai Park
Campaign	Mixed	Summer 2026
Tag	Various	+ Family

Confirmation

Show:

- 186 assets will be updated.

Processing

For large operations:

Queued

Processing

Completed

Failed

Failure Handling

Example:

182 successful

4 failed

Users should be able to:

- View Failed
- Retry

Audit

Bulk changes must maintain the same auditability as individual changes.

Screen 10 — Asset Activity, Recent Assets & Library Health

Purpose

Give administrators and DAM managers visibility into the health and operational quality of the asset library.

Library Health KPIs

Display:

- Assets Added
- Assets Updated
- Unclassified Assets
- Missing Metadata
- Orphaned Assets

- Archived Assets
- Duplicate Candidates
- Failed Processing
- Storage Growth

Recent Activity

Example:

Activity	Asset	User	Time
Uploaded	Summer_Hero.jpg	Sara	10:42
Metadata Updated	Venue_Map.pdf	Ahmed	10:38
Added to Collection	Promo_01.mp4	Leila	10:22
Archived	Old_Logo.ai	Admin	09:48

Library Health

Example:

- 96.8% Classified
- 98.4% Mandatory Metadata Complete
- 128 Unclassified
- 42 Missing Metadata
- 36 Duplicate Candidates

Storage Trend

Show:

- Used Storage
- Monthly Growth
- Assets Added

Orphaned Assets

TICVAI should identify assets that:

- are not in a collection,
- have incomplete classification,
- have no business context,
- or have not been used for a configured period.

This does **not mean** TICVAI should automatically delete them.

Actions

- Review Unclassified
- Fix Metadata
- Review Duplicate Candidates
- Review Orphaned Assets
- View Activity Log

Board 1 — Shared Configuration Requirements

Authorized administrators should be able to configure:

- Asset types
- supported file formats

- file-size limits
- upload rules
- asset naming rules
- folder hierarchy
- collection types
- category hierarchy
- subcategories
- metadata schemas
- mandatory metadata
- custom metadata fields
- taxonomy
- tag vocabulary
- tag governance
- asset statuses
- archive rules
- ownership fields
- language values
- search fields
- search filters
- library views
- bulk-operation permissions
- storage thresholds
- processing rules
- duplicate-warning behavior
- notifications

Critical Architecture — One Digital Asset Master

Every uploaded file should receive a permanent TICVAI:

Digital Asset ID

Example:

DAM-IMG-008421

This becomes the master reference.

Architecture:

Physical File



Digital Asset Master Record



Metadata



Versions



Rights



Collections



Usage / Distribution

Other TICVAI modules should reference the **Digital Asset ID**, rather than copying and independently managing the same media.

Critical Architecture — Folder vs Collection

Folder

Answers:

Where is the asset organized?

Collection

Answers:

Which business group does this asset belong to?

Example:

Dubai_Park_Hero.jpg

may physically belong to:

Marketing / Dubai Park / 2026

but simultaneously appear in:

Summer Campaign

Website Homepage

Approved Brand Assets

B2C Launch

without creating four copies.

Critical Architecture — DAM vs Document Management

The DAM should primarily manage **reusable digital content and media**.

Examples:

Images

Videos

Audio

Brand Assets

Creative Files

Venue Media

Ticket Media

Campaign Assets

It may store documents where required, but it should not unintentionally become a replacement for every operational document repository in TICVAI.

For example, a maintenance certificate linked to an asset may remain a maintenance document rather than becoming a marketing DAM asset.

Critical Architecture — DAM vs Ticket Media

TICVAI already has a **Ticket Media module**.

The recommended relationship is:

DAM

stores reusable media:

Logo

Background

Artwork

Campaign Image

Brand Asset

↓

Ticket Media

uses those assets to construct:

Ticket Templates

PDF Tickets

Wallet Passes

Email Attachments

Print Layouts

Therefore Ticket Media should be able to:

Select from DAM

rather than maintaining unnecessary duplicate files.

Critical Architecture — DAM vs Marketing & CRM

Marketing & CRM should manage:

Campaign

Audience

Communication

Journey

Promotion

while DAM manages:

Creative Asset

Therefore:

Marketing Campaign

↓

Select Approved DAM Assets

↓

Email / App / Web / Social / Campaign Usage

This creates a clean separation between **content management** and **campaign execution**.

Asset Status Model

I recommend a controlled asset lifecycle:

Uploading

↓

Processing

↓

Draft

↓
Active

↓
Archived

with additional states where needed:

Processing Failed

Restricted

Expired

Board 3 will expand governance around approval, rights and expiration.

Multi-Tenant Architecture

Every asset must inherit a clear data scope:

Tenant

↓
Venue / Business Unit

↓
Asset

TICVAI should support:

Tenant Private

Venue Specific

and later, through Board 3 governance:

Shared Across Authorized Venues

A user from Tenant A must never discover or access Tenant B's private DAM assets through search, collections, URLs, AI, APIs, or reports.

Search Architecture

Structured discovery should index:

Filename

-

Title

-

Description

-

Metadata

-

Taxonomy

-

Tags

-

Keywords

-

Collections

Board 2 can later add:

AI-generated labels

-

Image/video understanding

-

Semantic vectors

giving TICVAI:

Structured Search + Semantic Search

Audit Trail

All important actions should record:

Who

What

Asset

Before

After

When

Source

Examples:

- Uploaded asset
- changed metadata
- moved asset
- changed category
- added tag
- removed tag
- added to collection
- archived
- restored
- bulk updated

Detailed sharing/download/rights audit will be expanded in Board 3.

RBAC & Governance

Permissions should distinguish:

View DAM

Upload Asset

Edit Metadata

Manage Tags

Manage Taxonomy

Create Folder

Create Collection

Move Asset

Bulk Manage

Download Original

Archive Asset

Restore Asset

Manage DAM Configuration

View Activity

Permissions must respect both:

Role

and:

Tenant/Venue Data Scope

Important Addition — Source vs Rendition

Board 1 should recognize that an asset has an:

Original / Source File

Example:

Dubai_Hero_Master.psd

while multiple derived files may later exist:

Web JPG

Mobile WebP

Thumbnail

Instagram 1080×1350

Kiosk 1920×1080

These derived versions should **not be treated as unrelated digital assets by default**.

Board 2 should manage the detailed **rendition/transformation engine**.

This will become particularly important when TICVAI distributes the same content to B2C, kiosks, POS, apps and marketing channels.

Board 1 — Acceptance Criteria

Board 1 is accepted when TICVAI can successfully:

**Upload → Validate → Register → Classify → Organize → Tag → Search → Discover → Preview → Bulk
Manage → Audit**

digital assets across multiple tenants and venues.

The platform must preserve the relationship:

**Digital Asset ID → Original File → Metadata → Taxonomy → Folder → Collections → Tags → Business
Context → Activity History**

and provide a scalable foundation for the next boards without duplicating their responsibilities.

Board Boundary

Board 1 — Digital Asset Library & Organization

Asset ingestion, master record, metadata, taxonomy, folders, collections, tags, search and library operations.

Board 2 — AI Asset Intelligence & Versioning

AI classification, semantic/visual search, duplicate intelligence, versions, transformations and renditions.

Board 3 — Asset Governance, Rights & Sharing

Approvals, permissions, copyrights, usage rights, expiration, controlled sharing and audit governance.

Board 4 — Distribution, Analytics & Integration

Channel delivery, APIs, CDN/delivery links, usage, integrations, reporting and AI insights.

Board 1 = 10 Screens

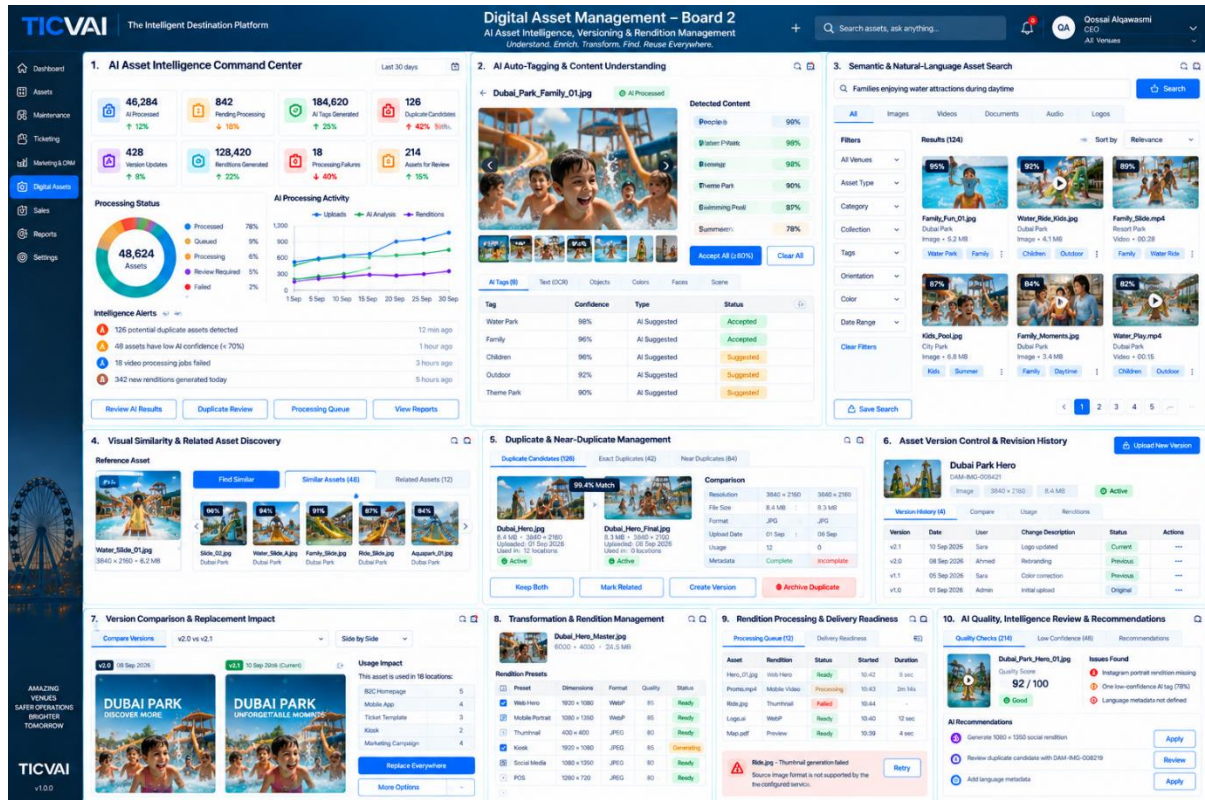
This gives development team a clear foundation: **Board 1 builds the library; the following boards make that library intelligent, governed and distributable.**

Module 23 — Digital Asset Management (DAM)

Board 2 — AI Asset Intelligence, Versioning & Rendition Management

Board Objective

Build the **intelligence and media-processing layer** of TICVAI DAM so the platform can automatically understand digital content, improve discovery, identify duplicates, manage versions, generate optimized renditions, and maintain relationships between source assets and their derivatives.



Board 1 answers:

Where is the asset, and how is it organized?

Board 2 answers:

What is inside the asset, how is it related to other assets, which version is current, and which format should be used?

The core architecture should be:

Original Asset → AI Understanding → Version Control → Transformation → Renditions → Intelligent Discovery

Screen 1 — AI Asset Intelligence Command Center

Purpose

Provide DAM administrators and content teams with an overview of AI processing, version activity, duplicate detection, rendition generation, and asset-quality issues.

KPI Cards

Display:

- AI-Processed Assets
- Pending AI Processing
- AI Tags Generated
- Duplicate Candidates
- Version Updates
- Renditions Generated
- Processing Failures
- Assets Requiring Review

Example:

KPI	Value
AI Processed	46,284
Pending	842
AI Tags	184,620
Duplicate Candidates	126
Version Updates	428
Renditions	128,420
Processing Failed	18
Review Required	214

Processing Health

Show:

- Processed
- Queued
- Processing
- Review Required
- Failed

AI Confidence Distribution

Show:

- 95–100%
- 80–94%
- 60–79%
- Below Threshold

Intelligence Alerts

Examples:

- ⚠ 126 potential duplicate assets detected.
- 🔴 48 assets contain AI classifications below the configured confidence threshold.
- 🔴 18 video-processing jobs failed.
- 🟡 342 new renditions were generated today.

Quick Actions

- Review AI Results
- Duplicate Review
- Version Activity
- Rendition Queue
- Processing Failures

Screen 2 — AI Auto-Tagging & Content Understanding

Purpose

Automatically analyze uploaded assets and generate useful descriptive metadata.

Asset Analysis

Example:

Dubai_Park_Family_01.jpg

AI detects:

People

Family

Children

Water Attraction

Outdoor

Daytime

Theme Park

Swimming Pool

AI-Generated Tags

Display each tag with confidence:

Suggested Tag	Confidence	Status
Water Park	98%	Accepted
Family	96%	Suggested
Children	94%	Suggested
Outdoor	92%	Suggested
Summer	78%	Review

Actions

- Accept
- Reject
- Edit
- Accept All Above Threshold

Supported Intelligence

Depending on media type, architecture should support:

Object Recognition

Scene Recognition

Logo Detection

Text Recognition

Color Analysis

Image Classification

Video Scene Analysis

Audio/Video Transcription

Document Content Extraction

where supported by the configured AI service.

Governance

AI tags should be distinguishable from:

Human Tags

and:

Controlled Taxonomy

Example:

AI Suggested

versus:

Approved Metadata

This distinction is important.

Screen 3 — Semantic & Natural-Language Asset Search

Purpose

Allow users to search based on meaning rather than exact filenames or tags.

Board 1 provided structured search.

Board 2 adds:

Semantic Search

Example query:

“Families enjoying water attractions during daytime.”

TICVAI should return relevant assets even if none contain those exact words in the filename.

Natural-Language Queries

Examples:

“Show me outdoor family images from Dubai Park.”

“Find vertical videos suitable for Instagram.”

“Show images containing the ORYX logo.”

“Find pictures of children playing near a water attraction.”

Search Result

Each result should show:

- Thumbnail
- asset
- relevance score
- AI tags
- matching reason
- format
- dimensions

- venue

Explain Match

Example:

94% Match

Matched because:

Family

Water Attraction

Outdoor

Daytime

Search Modes

Support:

- **Keyword**
- **Semantic**
- **Natural Language**
- **Combined**

Security

AI search must respect exactly the same:

Tenant + Venue + RBAC + Rights

as normal DAM search.

AI must never expose an asset the user cannot normally access.

Screen 4 — Visual Similarity & Related Asset Discovery

Purpose

Allow users to find visually related or similar media.

Find Similar

From an existing asset:

Find Similar Images

Results

Show similarity percentage:

96%

91%

87%

etc.

Similarity Factors

May include:

- Visual composition
- objects
- scene
- colors
- subject
- image structure

- embeddings

Use Cases

Examples:

Find another image from the same photo shoot

Find alternative hero images

Locate near-duplicates

Find similar attraction photography

Related Asset Groups

TICVAI should distinguish:

Similar Asset

Different content with similar visual characteristics.

from:

Duplicate Candidate

Potentially the same content.

from:

Version

An intentionally managed revision of the same digital asset.

from:

Rendition

A generated format/size derived from the same source.

These must not be treated as the same concept.

Screen 5 — Duplicate & Near-Duplicate Management

Purpose

Prevent the DAM from becoming filled with unnecessary copies of identical or nearly identical content.

Duplicate Detection

Support:

- **Exact Duplicate**
- **Near Duplicate**
- **Visually Similar**

Detection Methods

Architecture may use:

- File hash
- perceptual hash
- metadata
- visual similarity
- filename patterns
- AI embeddings

Duplicate Review

Example:

Potential Duplicate Group #DG-00428

Dubai_Hero.jpg

versus:

Dubai_Hero_Final.jpg

Similarity:

99.4%

Comparison View

Show:

Attribute	Asset A	Asset B
Resolution	3840×2160	3840×2160
File Size	8.4 MB	8.3 MB
Uploaded	01 Sep	08 Sep
Usage	12 locations	0 locations
Rights	Approved	Unknown

Actions

- Keep Both
- Mark Related
- Create Version Relationship
- Replace Duplicate
- Archive Duplicate

Critical Safety

TICVAI must show where each asset is currently used **before allowing consolidation or archive**.
Never automatically delete a duplicate simply because AI says it is similar.

Screen 6 — Asset Version Control & Revision History

Purpose

Maintain controlled versions of the same logical digital asset without creating unrelated master records.

Example

DAM-IMG-008421

Dubai Park Hero

Versions:

- v1.0 — Original
- v1.1 — Color Correction
- v2.0 — New Branding
- v2.1 — Approved Final

Version Table

Version	Date	User	Change	Status
v2.1	10 Sep	Sara	Logo updated	Current
v2.0	08 Sep	Ahmed	Rebranding	Previous

Version	Date	User	Change	Status
v1.1	05 Sep	Sara	Color correction	Previous
v1.0	01 Sep	Admin	Initial upload	Original

Version Actions**Upload New Version****Compare Versions****Restore Version****Download Version****View Usage****Current Version**

Only one version should normally be:

Current

Previous versions remain available according to retention policy.

Important**Uploading:**

Dubai_Hero_v2.jpg

should allow the user to choose:

New Asset

or:

New Version of Existing Asset

This prevents unnecessary duplication.

Screen 7 — Version Comparison & Replacement Impact**Purpose**

Allow users to compare versions and understand the consequences of making a new version current.

Comparison**Support:**

- **Side-by-Side**
- **Overlay**

Metadata Comparison

where appropriate.

Example**v2.0 vs v2.1****Changes:**

- New logo
- Updated campaign text
- dimensions unchanged
- file size changed
- metadata updated

Usage Impact

Before replacing current version:

This asset is currently used in **18 locations**.

Examples:

B2C Homepage

Mobile App

Ticket Template

Kiosk

Marketing Campaign

Replacement Options

Depending on integration:

Replace Everywhere

Replace Selected Uses

Keep Existing Uses on Old Version

Schedule Replacement

Board 4 will own detailed distribution and propagation.

Board 2 provides the version relationship and impact visibility.

Rollback

If an approved new version causes a problem:

Restore Previous Current Version

subject to permission.

All changes must be audited.

Screen 8 — Transformation & Rendition Management

Purpose

Generate channel-appropriate media from a master asset without forcing users to manually upload many independent copies.

Source Asset

Example:

Dubai_Hero_Master.jpg

6000 × 4000

Renditions

Generate:

Rendition	Dimensions	Format	Use
Web Hero	1920×1080	WebP	B2C
Mobile Hero	1080×1350	WebP	Mobile
Thumbnail	400×400	JPEG	DAM
Kiosk	1920×1080	JPEG	Kiosk
Social Portrait	1080×1350	JPEG	Social

Rendition	Dimensions	Format	Use
POS	1280×720	JPEG	POS

Transformation Operations

Support where applicable:

- Resize
- crop
- fit
- compress
- format conversion
- quality optimization
- thumbnail generation

Presets

Administrators should configure:

- **Rendition Presets**

Example:

B2C Hero

1920 × 1080 / WebP / Quality 85

Generation

Renditions can be:

Automatic

or:

On Demand

Important

A rendition should maintain:

Parent Asset ID

and:

Source Version

so TICVAI knows exactly where it came from.

Screen 9 — Rendition Processing & Delivery Readiness

Purpose

Monitor media-processing jobs and ensure required formats are ready before content is distributed.

Processing Queue

Asset	Rendition	Status	Started	Duration
Hero_01	Web Hero	Ready	10:42	8 sec
Promo.mp4	Mobile Video	Processing	10:43	2m 14s
Ride.jpg	Thumbnail	Failed	10:44	—

Status

Queued

Processing

Ready

Failed

Retrying

Failure Details

Example:

Source video codec is not supported by the configured transcoding service.

Actions

- **Retry**
- **Cancel**
- **View Error**
- **Regenerate**

Delivery Readiness

For each asset:

Web

 Ready

Mobile

 Ready

Kiosk

 Ready

Social

 Rendition Missing

This becomes important before Board 4 distributes content.

Processing Priority

Support:

- **Normal**
- **High**

Urgent

where permitted.

Screen 10 — AI Quality, Intelligence Review & Recommendations

Purpose

Provide a governed review workspace for AI results and identify content-quality issues before assets are reused or distributed.

AI Quality Checks

Depending on media type, detect or flag:

- Low resolution
- poor orientation for intended channel
- excessive compression
- blurry imagery
- duplicate content
- missing metadata

- missing classification
- inconsistent tags
- unsuitable dimensions
- missing required renditions

Example

Dubai_Park_Hero_01.jpg

Quality Score

92 / 100

 **Good**

Issues:

- Instagram portrait rendition missing
- one low-confidence AI tag
- metadata language not defined

Review Queue

Asset	Issue	Severity	AI Confidence	Status
Hero_01	Missing Rendition	Medium	100%	Open
Ride_08	Duplicate Candidate	High	98%	Review
Promo_04	Low Resolution	High	96%	Open
Venue_01	Tag Conflict	Low	72%	Review

AI Recommendations

Examples:

“Generate 1080×1350 social rendition.”

“Review duplicate relationship with DAM-IMG-008219.”

“Replace tag ‘Waterpark’ with controlled tag ‘Water Park.’”

“This image may be too low resolution for kiosk display.”

Actions

- **Accept Recommendation**
- **Reject**
- **Review Asset**
- **Generate Rendition**
- **Resolve Issue**

Board 2 — Shared Configuration Requirements

Authorized administrators should configure:

- AI processing enabled/disabled
- AI provider/model configuration
- supported AI capabilities
- confidence thresholds
- auto-tag thresholds
- human-review thresholds
- approved AI tag categories

- semantic-search configuration
- visual similarity thresholds
- exact duplicate rules
- near-duplicate thresholds
- duplicate review workflow
- version numbering rules
- version retention
- current-version rules
- version replacement permissions
- rendition presets
- image dimensions
- supported output formats
- compression/quality settings
- crop/fit rules
- video rendition profiles
- processing priorities
- retry rules
- processing timeout
- quality thresholds
- quality-review rules
- notification rules

Critical Architecture — Four Different Relationships

1. Version

Same logical asset, intentionally revised.

Hero v1 → Hero v2

2. Rendition

Same version transformed for a channel.

Hero v2 → WebP 1920×1080

3. Duplicate

Unnecessary additional copy.

Hero_Final.jpg ≈ Hero_Final_Copy.jpg

4. Related Asset

Different asset with a meaningful relationship.

Campaign Hero ↔ Campaign Video

The data model should support all four relationships independently.

Critical Architecture — Master Asset / Version / Rendition

Recommended hierarchy:

Digital Asset

DAM-IMG-008421

↓

Versions

- v1.0
- v2.0
- v2.1 CURRENT



Renditions of v2.1

- Original**
- Web Hero**
- Mobile**
- Thumbnail**
- Kiosk**

Social

This avoids creating six unrelated records for one piece of content.

Critical Architecture — AI Metadata Must Be Traceable

Every AI-generated result should store:

AI Result

-

Confidence

-

Provider / Model Reference

-

Processing Date

-

Human Review Status

Example:

Tag: Water Park

Source: AI

Confidence: 98%

Status: Approved

If a human changes it:

Status: Human Modified

This is particularly important as TICVAI may change AI providers over time.

Critical Architecture — AI Provider Independence

Architecture:

TICVAI DAM



TICVAI AI Service Layer



Configured AI Provider / Model

This allows TICVAI to use different providers for:

- image understanding,
- OCR,
- transcription,
- embeddings,
- moderation/classification,
- or future proprietary models.

It also supports your broader TICVAI strategy of controlling AI consumption and pricing.

Critical Architecture — Search Index

Board 1 structured metadata and Board 2 AI intelligence should feed one discovery layer:

Human Metadata

-

Controlled Taxonomy

-

Human Tags

-

AI Tags

-

Extracted Text

-

Transcription

-

Semantic Embeddings



TICVAI Asset Search Index

This allows users to use one search interface rather than separate “normal search” and “AI search” databases.

Critical Architecture — Source Preservation

The original uploaded source must normally remain immutable.

For example:

Original

6000 × 4000 TIFF

should not be overwritten when TICVAI generates:

1920 × 1080 WebP.

Instead:

Original Source



Version



Transformation



Rendition

This preserves quality and traceability.

Critical Architecture — Version Propagation

When a new version becomes current, TICVAI should **not blindly replace every live use**.

Different consumers may have different policies:

Auto-Follow Current Version

Example:

Internal DAM thumbnail.

Approval Required

Example:

B2C homepage.

Version Locked

Example:

Historical campaign or already-issued ticket.

Therefore usage references should support:

Follow Current

or:

Pin Version

This becomes essential in Board 4.

AI Cost & Consumption Control

Because TICVAI is multi-tenant, AI processing should be measurable.

Track:

- Assets processed
- images analyzed
- video minutes analyzed
- audio minutes transcribed
- OCR pages
- embedding generation
- semantic searches
- AI requests
- estimated/actual AI consumption cost

This allows future TICVAI subscription plans to include:

AI Usage Quotas

or:

Consumption-Based AI Pricing

without redesigning DAM later.

AI Governance

AI can:

- suggest tags,
- classify assets,

- extract text,
- transcribe media,
- identify visual similarity,
- detect duplicate candidates,
- recommend renditions,
- identify quality issues,
- improve search,
- suggest related assets.

AI should not autonomously:

- delete assets,
- archive business-critical assets,
- merge duplicates,
- replace current versions,
- publish assets,
- approve rights,
- override taxonomy governance,
- or distribute content externally.

These remain governed actions.

RBAC & Governance

Permissions should distinguish:

Use AI Search

Review AI Metadata

Approve AI Tags

Manage Duplicate Candidates

Create New Version

Make Version Current

Restore Version

Compare Versions

Create Rendition

Manage Rendition Presets

Regenerate Rendition

View Processing Errors

Manage AI Configuration

View AI Consumption

View AI Cost

All sensitive changes should be audited.

Board 2 — Acceptance Criteria

Board 2 is accepted when TICVAI can successfully:

**Analyze → Understand → Auto-Classify → Search Semantically → Find Similar → Detect Duplicates →
Manage Versions → Compare → Transform → Generate Renditions → Review Quality**

while preserving the relationship:

**Digital Asset → AI Intelligence → Version → Transformation → Rendition → Search Index → Usage
Readiness**

and maintaining full tenant isolation, permissions, traceability, and human governance.

Board Boundary

Board 1 — Digital Asset Library & Organization

Master library, ingestion, metadata, taxonomy, folders, collections and structured search.

Board 2 — AI Asset Intelligence, Versioning & Renditions

AI understanding, semantic discovery, duplicate intelligence, versions, transformations and media processing.

Board 3 — Asset Governance, Rights & Sharing

Approval, ownership, rights, expiration, permissions, controlled sharing and compliance.

Board 4 — Distribution, Analytics & Integration

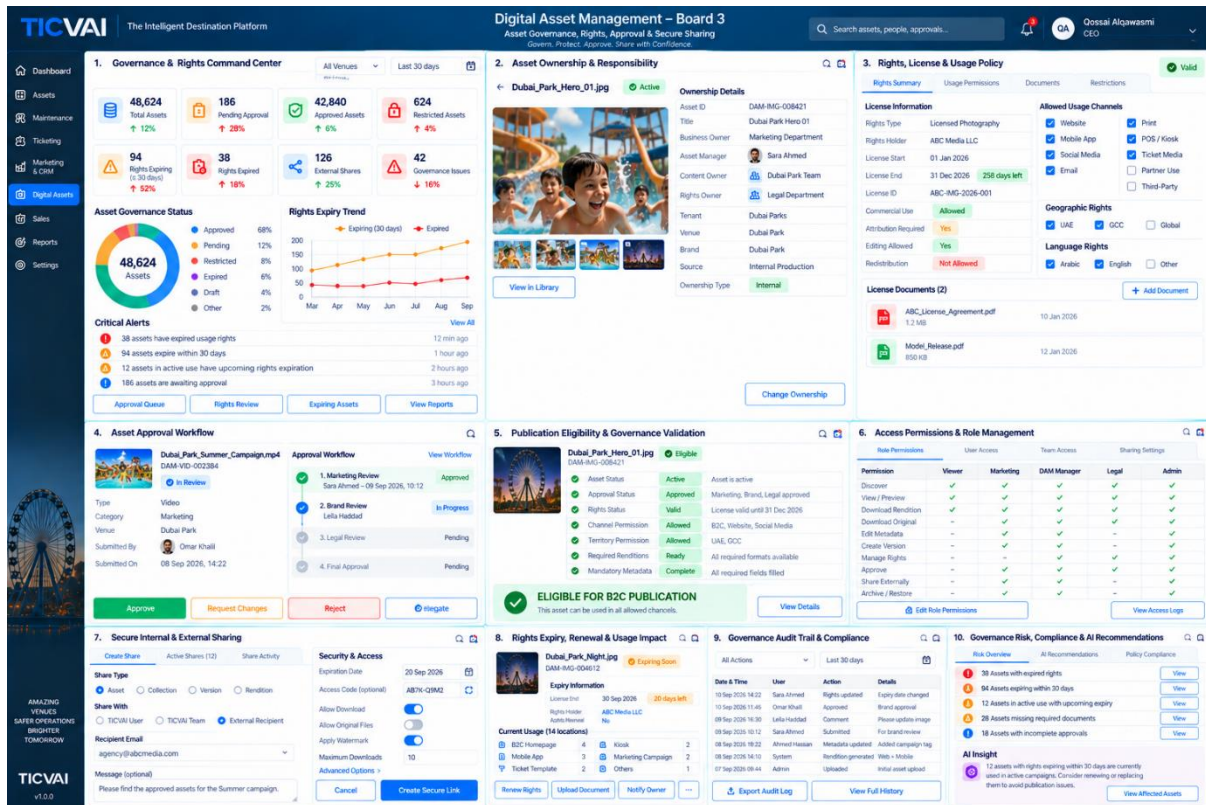
Delivery, channel distribution, APIs, CDN, usage analytics, replacement propagation and integrations.

Board 2 = 10 Screens.

Board 3 — Asset Governance, Rights, Approval & Secure Sharing

Board Objective

Build the governance and control layer of TICVAI DAM so every digital asset can be governed throughout its lifecycle based on ownership, approval status, usage rights, licensing terms, expiration, tenant/venue permissions, channel restrictions, sharing policies, and audit requirements.



Board 1 established the asset library. Board 2 added intelligence, versions, and renditions. Board 3 determines:

Who can access the asset, who can approve it, where it may be used, for how long, and whether it can safely be shared or published.

The core architecture should be:

Asset → Ownership → Rights → Approval → Permission → Usage Eligibility → Sharing → Expiration → Audit

Screen 1 — DAM Governance & Rights Command Center

Purpose

Provide DAM administrators, brand managers, legal/compliance teams, and content managers with one overview of asset governance.

KPI Cards

Display:

- Active Assets
- Pending Approval

- **Approved Assets**
- **Restricted Assets**
- **Rights Expiring Soon**
- **Rights Expired**
- **External Shares**
- **Governance Issues**
- **Approval SLA Breaches**
- **Assets Blocked from Distribution**

Example:

KPI	Value
Governed Assets	48,624
Pending Approval	186
Approved	42,840
Restricted	624
Rights Expiring	94
Rights Expired	38
External Shares	126
Governance Issues	42

Governance Health

Show:

- **Approved**
- **Pending**
- **Restricted**
- **Expired**
- **Blocked**
- **Critical Alerts**

Examples:

- 38 assets have expired usage rights.
- 94 asset licenses expire within 30 days.
- ⚠ 12 assets currently used by active channels have upcoming rights expiration.
- 186 assets are awaiting approval.

Quick Actions

- **Approval Queue**
- **Rights Review**
- **Expiring Assets**
- **External Shares**
- **Governance Issues**

Audit

Screen 2 — Asset Ownership & Responsibility Management

Purpose

Define who owns and is responsible for each digital asset from a business and operational perspective.

This should be separate from the person who simply uploaded the file.

Asset Ownership Record

Example:

Dubai Park Summer Hero

Asset ID: DAM-IMG-008421

Business Owner: Marketing Department

Asset Manager: Sara Ahmed

Tenant: Dubai Parks

Venue: Dubai Park

Brand: Dubai Park

Source: Internal Production

Ownership Types

Support:

- Internal
- Agency
- Photographer
- Vendor
- Partner
- Licensed Third Party
- Client Supplied
- Responsibility Roles

Allow assignment of:

Business Owner

Content Owner

Rights Owner

Asset Manager

Approver

Owner Change

Support controlled ownership transfer.

Example:

Previous Owner: Campaign Team

↓

New Owner: Corporate Marketing

Reason

Campaign completed — transferred to corporate library.

Important

Changing the business owner must not change the creator, uploader, or historical ownership records.

Screen 3 — Rights, License & Usage Policy Management**Purpose**

Define exactly how an asset is legally and commercially allowed to be used.

Rights Profile

Example:

DAM-IMG-008421

Rights Status:  **Valid**

Rights Type: Licensed Photography

Rights Holder: ABC Media LLC

License Start: 01 Jan 2026

License End: 31 Dec 2026

Usage Rights

Configure:

Internal Use

Website

Mobile App

Social Media

Email

Digital Advertising

Print

POS

Kiosk

Ticket Media

Partner Distribution

Third-Party Distribution

Geographic Rights

Example:

✓ **UAE**

✓ **GCC**

✗ **Global**

Language Rights

Example:

✓ **Arabic**

✓ **English**

Commercial Restrictions

Support:

- **Commercial use permitted**
- **Non-commercial only**

- Attribution required
- Editing permitted
- Cropping permitted
- Redistribution permitted
- Third-party use prohibited
- Rights Documents

Attach:

- License agreement
- model release
- photographer agreement
- agency agreement
- consent
- copyright document
- other supporting evidence

Rights Status

- ☒ Valid
- ☒ Expiring Soon
- ☒ Expired
- ☒ Restricted
- ☐ Not Defined

Screen 4 — Asset Approval Workflow Management

Purpose

Provide configurable governance before an asset becomes approved for operational or public use.

Approval Workflow

Example:

Draft



Submitted



Marketing Review



Brand Review



Legal Review



Approved

Workflow Configuration

Approval workflow should be configurable by:

Asset Type

Category

Tenant

Venue

Brand

Intended Usage**Risk Level****Example**

A normal internal image may require:

Marketing Approval

while a public campaign containing third-party licensed content may require:

Marketing → Brand → Legal

Approval Queue

Asset	Type	Submitted	Stage	Owner	SLA
Summer Hero	Image	10 Sep	Brand	Sara	3h
Promo Video	Video	10 Sep	Legal	Omar	7h
Venue Map	PDF	09 Sep	Operations	Ahmed	Overdue

Actions

- Approve
- Reject
- Request Changes
- Delegate
- Add Comment
- Rejection

Require:**Reason**

and optionally:

Required Changes**Screen 5 — Publication Eligibility & Governance Validation****Purpose**

Determine whether an asset is actually eligible to be used or distributed.

This screen is particularly important.

An asset being technically available in DAM does not mean it should be publishable.

Eligibility Check**Example:**

Dubai Park Hero

Asset Status:

☒ Active

Approval:

☒ Approved

Rights:

☒ Valid

Required Rendition:

☒ Ready

Mandatory Metadata:

☒ Complete

Channel Permission:

☒ B2C Allowed

Overall:

☒ ELIGIBLE FOR B2C

Another example:

Summer Promo Video

Approval:

☒ Approved

Rights:

☐ Expired

Overall:

☒ BLOCKED

Reason:

Usage rights expired on 01 September 2026.

Validation Rules

Before distribution, TICVAI should check:

Asset Active

-

Approved

-

Rights Valid

-

Channel Permitted

-

Territory Permitted

-

Rendition Available

-

Mandatory Metadata Complete

=

Distribution Eligible

Board 4 should consume this eligibility result before publishing.

Screen 6 — Role-Based Asset Access & Permission Management

Purpose

Control who can discover, view, edit, download, share, approve, or manage assets.

Access Scope

- Support:
- Tenant
- ↓
- Venue
- ↓
- Folder / Collection
- ↓
- Asset
- Permission Types
- Distinguish:
- Discover
- View
- Preview
- Download Rendition
- Download Original
- Edit Metadata
- Create Version
- Share
- Approve
- Manage Rights
- Archive
- Example Role Matrix

Permission	Viewer	Marketing	DAM Manager	Legal	Admin
View	✓	✓	✓	✓	✓
Download	✓	✓	✓	✓	✓
Edit	—	✓	✓	—	✓
Approve	—	✓	✓	✓	✓
Rights	—	—	✓	✓	✓
Share	—	✓	✓	—	✓

- Sensitive Assets
- Support additional classification such as:
- Public
- Internal
- Confidential
- Restricted
- Important Security Rule

- If the user does not have permission to discover an asset, the asset should not appear in:
- DAM search
 - AI search
 - collections
 - recommendations

- related assets
- API results

Security must be enforced before retrieval, not only when opening the file.

Screen 7 — Secure Internal & External Sharing

Purpose

Allow controlled sharing without users manually downloading assets and sending uncontrolled copies.

Share Asset / Collection

Allow sharing:

Asset

Version

Rendition

Collection

Share With

TICVAI User

TICVAI Team

Authorized External Recipient

External Share Configuration

Capture:

Recipient

Expiration Date

Password / Access Code

Download Allowed

Original Download Allowed

Watermark Required

Maximum Downloads

Access Region/IP Rules where configured

Example

Summer Campaign Collection

External Share:

Agency ABC

Expires:

20 Sep 2026

Permission:

View + Download Approved Renditions

Original Files:

Not Allowed

Share Status

● Active

● Expiring

- Expired
- Revoked

Actions

- Copy Secure Link
- Extend
- Change Permission
- Revoke
- View Activity

Screen 8 — Rights Expiry, Renewal & Usage Impact

Purpose

Prevent expired assets from continuing to be used across TICVAI channels.

Expiry Dashboard

Show:

- Expires in 7 Days
- Expires in 30 Days
- Expires in 60 Days
- Expired
- Example
- DAM-IMG-008421

Rights expire:

30 Sep 2026 — 20 Days

Current Usage:

B2C Homepage

Mobile App

Marketing Campaign

Kiosk

Impact

⚠ This asset is currently used in 14 active locations.

Actions

- Renew Rights
- Upload Renewal Document
- Change Expiry
- Find Replacement
- Notify Owner
- Expiry Workflow

Example:

60 Days

Owner notification



30 Days

Owner + Manager



7 Days

Escalation



Expired

Distribution restriction based on configured policy

Important

TICVAI should not simply remove an expired image from a live website without understanding the configured replacement behavior.

Board 4 should control actual channel replacement.

Board 3 determines:

No Longer Eligible

Screen 9 — Governance Audit Trail & Compliance Evidence

Purpose

Provide complete traceability for asset governance.

Audit Timeline

Capture:

Asset Uploaded



Metadata Updated



AI Tags Reviewed



Rights Added



Submitted for Approval



Brand Approved



Legal Approved



Shared Externally



Rights Renewed

Audit Table

Time	User	Action	Before	After	Source
10:42	Sara	Rights Updated	30 Sep	31 Dec	DAM
10:38	Omar	Approved	Pending	Approved	Workflow
10:12	Admin	Share Revoked	Active	Revoked	DAM

Filters

Support:

- User
- Action

- Asset
- Date
- Tenant
- Venue
- Governance Type
- Compliance Evidence

Authorized users should be able to view:

- Approval history
- license evidence
- ownership history
- sharing activity
- permission changes
- expiration changes
- version approval history

Audit Principle

Audit history must be append-only from the normal user experience.

Users should not be able to rewrite history.

Screen 10 — Governance Risk, Compliance & AI Recommendations

Purpose

Provide a consolidated view of governance risks and use AI to help identify problems requiring human attention.

Governance Risk KPIs

Display:

- Assets Without Rights
- Expired Assets
- Rights Expiring
- Unapproved Assets
- Restricted Assets in Active Usage
- External Shares
- Long-Lived Shares
- Missing Evidence
- Approval SLA Breaches
- Risk Queue

Asset	Risk	Usage	Severity	Action
Summer Hero	Rights Expiring	14	High	Review
Promo Video	Missing Release	4	Critical	Block
Old Logo	Outdated Brand	8	High	Replace
Campaign PDF	Approval Pending	2	Medium	Escalate

AI Governance Insights

Examples:

“12 assets with rights expiring within 30 days are currently used in active campaigns.”

“Three external shares have remained active longer than the tenant's normal 30-day sharing policy.”

“Promo_Video_04 is missing a model-release document required by the configured governance policy.”

“Eight channels are still referencing an older brand asset version.”

Recommended Actions

Review Rights

Request Approval

Find Replacement

Revoke Share

Request Documentation

Escalate Approval

Human Governance

All recommendations must require an authorized user to take the actual governed action.

Board 3 — Shared Configuration Requirements

Authorized administrators should configure:

- **Ownership types**
- **responsibility roles**
- **rights types**
- **rights statuses**
- **license types**
- **usage channels**
- **geographic rights**
- **language rights**
- **commercial restrictions**
- **attribution requirements**
- **editing/cropping permissions**
- **redistribution rules**
- **mandatory rights documents**
- **rights-expiry thresholds**
- **renewal workflows**
- **approval workflows**
- **approval stages**
- **approver roles**
- **approval SLA**
- **rejection reasons**
- **publication-eligibility rules**
- **asset sensitivity levels**
- **access permissions**
- **folder/collection permissions**
- **download permissions**
- **original-file permissions**
- **external-sharing policies**
- **external-share expiration**

- password/access-code requirements
- maximum downloads
- watermark requirements
- audit retention
- governance-risk thresholds
- notifications
- escalations

Critical Architecture — Approval Is Not Rights

An asset can be:

Approved

but:

Not Legally Usable

Example:

Marketing approves a photograph.

However:

License expired.

Therefore:

Approval = Approved

but:

Rights = Expired

Result:

● DISTRIBUTION BLOCKED

Likewise, an asset may have valid legal rights but still be waiting for brand approval.

Both conditions must be evaluated independently.

Critical Architecture — Rights at Asset and Version Level

Rights should normally be associated with the logical asset, but TICVAI must support version-specific differences where necessary.

Example:

v1

Old photography with license expiring December.

v2

New photography owned permanently by the client.

Therefore the data model must be able to determine:

Which rights apply to the exact version being distributed?

Critical Architecture — Usage Rights vs Access Permissions

These are also different.

Access Permission

Answers:

Can Qossai see/download this asset inside TICVAI?

Usage Rights**Answers:**

Is TICVAI legally/business-wise allowed to use this asset on Instagram?

A user may have permission to view an asset while the asset itself is not permitted for external publication.

Critical Architecture — Publication Eligibility Engine

I strongly recommend implementing a reusable:

Asset Eligibility Engine

Rather than every TICVAI module independently checking DAM governance.

Example request from Ticket Media:

“Can DAM-IMG-008421 v2.1 be used for a B2C ticket in UAE today?”

DAM returns:

ELIGIBLE

or:

BLOCKED

with reason codes:

RIGHTS_EXPIRED

APPROVAL_REQUIRED

CHANNEL_NOT_ALLOWED

TERRITORY_NOT_ALLOWED

ASSET_RESTRICTED

RENDITION_NOT_READY

MANDATORY_METADATA_MISSING

This same engine can serve:

B2C

Mobile App

Ticket Media

Marketing

POS

Kiosk

Digital Signage

and future integrations.

Critical Architecture — Governance Inheritance

TICVAI should support controlled inheritance.

Example:

Collection

Summer Campaign

Rights policy:

UAE only

Website + Social

Assets may inherit those rules where configured.

But if an individual asset has stricter rights:

Asset Restriction

Social only

the stricter rule should win.

Recommended principle:

The most restrictive applicable governance rule wins unless an authorized explicit override is permitted.

Critical Architecture — Secure Sharing

External sharing should never expose the underlying private storage location.

Architecture:

Recipient



Secure TICVAI Share



Authorization / Token Validation



Permission Check



Rights Check



Requested Rendition



Controlled Delivery

The public-facing token should be:

- revocable
- time-limited
- non-predictable
- auditable

and should not expose internal file-system or storage credentials.

Critical Architecture — Download Original vs Rendition

Board 2 established:

Original

and:

Renditions

Board 3 must apply separate permissions.

For example, an agency may be allowed:

✓ Download Web JPG

but:

✗ Download Original PSD

This is important for both security and intellectual-property protection.

Watermarking

For restricted external sharing, TICVAI should support configurable watermarking.

Examples:

CONFIDENTIAL

or:

Shared with Agency ABC — 10 Sep 2026

Watermarking should preferably apply to a generated delivery rendition, not permanently modify the master asset.

Expiration & Live Usage

This is one of the most important cross-board requirements.

Example:

Asset rights expire tonight.

Current usage:

- B2C homepage
- mobile app
- kiosk
- active campaign

Board 3 determines:

EXPIRES AT 23:59

and after expiry:

NOT ELIGIBLE

Board 4 should determine what happens operationally:

Use replacement

Remove content

Fallback asset

Hold publication

Notify channel owner

This separation avoids dangerous automatic behavior.

Personal Data & Consent Readiness

DAM may contain photography or video involving identifiable people.

The architecture should therefore allow an asset to reference:

Consent / Model Release

where the client's governance policy requires it.

Example:

Photo contains identifiable people

Required:

Model Release

Status:

 Valid

or:

 Missing

For children/minors, tenant policies may require additional consent evidence.

TICVAI DAM should support the governance relationship without attempting to become a full legal-consent platform.

AI Governance

AI can assist with:

- identifying missing rights metadata
- detecting assets likely containing identifiable people
- detecting old/outdated logos
- identifying possible governance inconsistencies
- prioritizing rights-expiry risk
- identifying long-lived external shares
- identifying missing approval steps
- recommending replacement assets

AI must not autonomously:

- approve an asset,
- grant legal rights,
- extend a license,
- interpret a contract as legally authoritative,
- publish restricted content,
- override rights,
- revoke business-critical content without policy,
- or approve external distribution.

RBAC & Segregation of Duties

Permissions should distinguish:

View Governance

Manage Ownership

Manage Rights

Upload Rights Evidence

Submit for Approval

Approve Marketing

Approve Brand

Approve Legal

Override Governance

Manage Permissions

Create External Share

Download Original

Revoke Share

Renew Rights

View Audit

Export Compliance Evidence

For sensitive workflows, the person uploading or editing the asset should not necessarily be able to perform the final approval.

Board 3 — Acceptance Criteria

Board 3 is accepted when TICVAI can successfully:

Assign Ownership → Define Rights → Attach Evidence → Submit → Review → Approve → Validate
 Eligibility → Control Access → Securely Share → Monitor Expiry → Block Invalid Usage → Audit →
 Identify Governance Risk

while preserving the governance chain:

Digital Asset → Version → Owner → Rights → Evidence → Approval → Permission → Eligibility → Sharing →
 Usage Impact → Expiration → Audit

The system must ensure that an asset being present in DAM does not automatically mean it can be published or externally distributed.

Board Boundary

Board 1 — Digital Asset Library & Organization

Library, ingestion, metadata, taxonomy, collections and structured discovery.

Board 2 — AI Asset Intelligence, Versioning & Renditions

AI understanding, semantic search, duplicates, versions, transformations and renditions.

Board 3 — Asset Governance, Rights, Approval & Secure Sharing

Ownership, licenses, rights, approvals, permissions, eligibility, sharing, expiration and governance audit.

Board 4 — Distribution, Analytics & Integration

Publishing/delivery, channel integration, CDN/API, usage tracking, propagation, analytics and integration monitoring.

Board 3 = 10 Screens.

Board 4 — Asset Distribution, Delivery, Integration & Analytics

Board Objective

Build the distribution and consumption layer of TICVAI DAM, allowing approved digital assets to be securely delivered and reused across TICVAI modules, websites, mobile applications, kiosks, POS, ticket media, marketing channels, digital signage, partner systems, and external applications.



Board 4 should also provide complete visibility into where each asset is being used, which version/rendition is being delivered, how replacements propagate, whether delivery is healthy, and how assets perform across channels.

The core architecture should be:

Approved Asset → Eligibility Validation → Channel → Rendition → Secure Delivery → Usage Reference
→ Monitoring → Analytics → Replacement

This board completes the end-to-end DAM lifecycle.

Screen 1 — Asset Distribution & Delivery Command Center

Purpose

Provide a centralized operational overview of asset distribution across TICVAI and connected external channels.

Global Filters

Support:

Tenant | Venue | Channel | Asset Type | Distribution Status | Date Range

KPI Cards

Display:

- **Assets in Active Use**
- **Active Asset References**
- **Delivery Requests**
- **Active Channels**
- **CDN/Data Delivery**
- **Failed Deliveries**
- **Assets Requiring Replacement**
- **Expiring Assets in Active Use**
- **API Requests**
- **Delivery Availability %**

Example:

KPI	Value
Assets in Active Use	18,426
Active References	84,218
Active Channels	28
Delivery Requests Today	1.84M
Delivery Availability	99.98%
Failed Deliveries	42
Replacement Required	18
Expiring in Active Use	12

Distribution by Channel**Show:**

- **B2C Website**
- **Mobile App**
- **POS**
- **Kiosk**
- **Ticket Media**
- **Marketing**
- **Digital Signage**
- **Partner/API**
- **Delivery Health**

Show:

-  **Healthy**
-  **Degraded**
-  **Failed**
-  **Inactive**
- **Critical Alerts**

Examples:

- 4 live asset references are currently failing delivery.
- 12 actively used assets have rights expiring within 30 days.
- ⚠ 6 channels are still referencing an outdated asset version.

18 replacement recommendations require review.

Quick Actions

- Distribution Map
- Channel Management
- Delivery Monitor
- Replacement Center
- API & Integration
- Analytics

Screen 2 — Asset Usage & Distribution Map

Purpose

Show exactly where a digital asset is currently being used.

This is one of the most important screens in the DAM.

Asset Example

Dubai Park Hero

Asset ID: DAM-IMG-008421

Current Version: v2.1

Active References: 18

Usage Map

Show:

- TICVAI DAM Asset
- ↓

B2C Website — 5

Mobile App — 4

Kiosk — 3

Ticket Media — 2

Marketing Campaign — 3

Digital Signage — 1

Usage Table

Consumer	Location	Version	Rendition	Status
B2C	Homepage	v2.1	Web Hero	Live
Mobile	Home Banner	v2.1	Mobile Hero	Live
Kiosk	Welcome Screen	v2.0	Kiosk	Outdated
Ticket Media	Annual Pass	v2.1	Ticket Image	Live
Campaign	Summer 2026	v2.1	Social	Live

Reference Type

Each usage should identify whether it is:

Follow Current Version

or:

Pinned Version

Drill Down

From the asset, user should be able to open:

Consumer Module

Usage Record

Version

Rendition

Distribution Status

Critical Requirement

This screen provides the answer to:

“If I change or archive this asset, what will be affected?”

Screen 3 — Channel & Distribution Configuration

Purpose

Configure how DAM content can be consumed by different TICVAI modules and external channels.

Channel Directory

Example:

Channel	Type	Assets	Delivery	Status
TICVAI B2C	Internal	4,284	CDN	Active
Mobile App	Internal	3,842	CDN/API	Active
POS	Internal	1,264	CDN	Active
Kiosk	Internal	842	CDN	Active
Ticket Media	Internal	2,184	Internal/API	Active
Partner Portal	External	624	API	Active

Channel Profile

Configure:

- Channel name
- channel type
- tenant
- venue scope
- allowed asset types
- allowed categories
- required approval
- required rights
- preferred rendition
- fallback rendition
- delivery method
- caching policy
- authentication requirement

Example

Mobile App

Preferred:

Mobile WebP — 1080 × 1350

Fallback:

Web JPEG

Rights Required:

Mobile App Usage

Approval Required:

Yes

Distribution Eligibility

Every channel request must call the Board 3 Asset Eligibility Engine.

Screen 4 — Secure Delivery URL, CDN & Rendition Delivery

Purpose

Provide optimized and controlled delivery of DAM assets without exposing private storage.

Delivery Record

Example:

DAM-IMG-008421

Version:

v2.1

Rendition:

Web Hero

Format:

WebP

Dimensions:

1920 × 1080

Delivery:

 Ready

Delivery Methods

Support architecture for:

CDN URL

Authenticated URL

Signed URL

API Response

Internal TICVAI Reference

Delivery Flow

Consumer Request



Asset Reference



Permission / Eligibility Check



Version Resolution



Rendition Resolution



CDN / Secure Delivery

URL Security

Support:

- signed URLs
- tokenized access
- configurable expiry
- authentication
- authorization
- cache policy
- revocation where applicable

Important

TICVAI should never expose the private storage path or storage credentials.

Rendition Selection

Example:

Request from Mobile:

DAM-IMG-008421

TICVAI resolves:

v2.1 → Mobile Hero → WebP

The consumer does not need to know the underlying file-storage structure.

Screen 5 — Asset Replacement & Propagation Management

Purpose

Safely manage replacement of assets that are outdated, expired, incorrect, rebranded, or otherwise no longer suitable.

Replacement Example

Old Asset

DAM-IMG-006214

Old Dubai Park Logo

Used in:

24 Active Locations

Replacement

DAM-IMG-009842

Dubai Park Logo 2026

Impact Table

Channel	Usage	Current	Replacement	Action
B2C	Header	Old Logo	New Logo	Replace
Mobile	Splash	Old Logo	New Logo	Replace
Kiosk	Home	Old Logo	New Logo	Approval
Ticket	PDF	Old Logo	New Logo	Keep Historical

Replacement Options

Support:

- Replace Everywhere Eligible
- Replace Selected Uses
- Schedule Replacement
- Keep Existing Version
- Create Fallback
- Important Historical Rule

A replacement should not modify historical outputs that should remain immutable.

Example:

An already-issued ticket PDF should not necessarily be regenerated because the logo changed.

Replacement Preview

Before confirmation:

18 live references will change.

4 historical references will remain unchanged.

Rollback

Where supported:

Rollback Replacement

with complete audit history.

Screen 6 — Fallback, Expiry & Distribution Continuity

Purpose

Ensure channels do not break when an asset becomes unavailable, expired, blocked, or fails delivery.

Distribution Problem

Example:

Summer Hero

Status:

● Rights Expired

Used by:

B2C Homepage

Configured Fallback

DAM-IMG-009211 — Default Dubai Park Hero

Resolution

Primary Asset

● Not Eligible



Fallback Asset

● Eligible



Deliver Fallback

Fallback Hierarchy

Support:

- **Usage-Level Fallback**

**Campaign Fallback****Channel Fallback****Venue Default****Tenant Default****Failure Reasons****Support:**

- **Asset archived**
- **rights expired**
- **approval revoked**
- **rendition unavailable**
- **asset restricted**
- **delivery failure**
- **version unavailable**

Actions

- **Assign Replacement**
- **Assign Fallback**
- **Notify Owner**
- **Pause Distribution**

Governance

TICVAI should never silently substitute an unrelated asset.

Fallback must be configured or governed.

Screen 7 — DAM API & Integration Hub

Purpose

Provide controlled APIs and integration services for internal TICVAI modules and approved external systems.

Integration Directory**Examples:**

TICVAI B2C

TICVAI Mobile

Ticket Media

Marketing & CRM

POS

Kiosk

Digital Signage

Partner Portal

External CMS

API Capabilities

Support APIs for:

Search Assets

Retrieve Asset Metadata

Retrieve Approved Renditions

Check Eligibility

Get Current Version

Get Delivery URL

Register Usage

Remove Usage Reference

Example Request

Consumer requests:

DAM-IMG-008421

Channel:

B2C

Purpose:

Homepage Hero

TICVAI returns:

Eligible: Yes

Version: v2.1

Rendition: Web Hero

Delivery: Secure CDN

Integration Authentication

Support appropriate:

- **API credentials**
- **OAuth/token mechanisms**
- **service accounts**
- **scopes**
- **tenant context**

API Governance

Consumers should receive only assets permitted by:

Tenant + Venue + Role/Scope + Rights + Eligibility

Screen 8 — Delivery Monitoring & Integration Health

Purpose

Monitor the technical health of DAM delivery and connected integrations.

Health KPIs

Display:

- **Delivery Availability**
- **Average Response Time**
- **CDN Cache Hit %**
- **API Success %**
- **Failed Requests**

- Processing Errors
- Integration Failures
- Broken Asset References
- Integration Health

Integration	Availability	Response	Errors	Status
B2C	99.99%	82ms	2	Healthy
Mobile	99.98%	94ms	4	Healthy
Kiosk	99.84%	126ms	18	Attention
Partner API	99.62%	184ms	28	Degraded

- Broken References

Detect:

Consumer references an asset/version that can no longer be resolved.

Example

Kiosk-04

Requested:

DAM-IMG-004218 / v1.2

Result:



VERSION_NOT_AVAILABLE

Actions

- View Consumer
- Assign Replacement
- Retry
- View Logs
- Notify Owner
- Alerting

Support alerts for:

- high error rate
- delivery latency
- CDN issues
- API failures
- repeated broken references
- eligibility failures
- unavailable renditions

Screen 9 — Asset Usage & Performance Analytics

Purpose

Show how digital assets are actually being consumed across TICVAI.

KPI Cards

Display:

- Assets Used
- Asset Requests
- Downloads

- External Shares
- Active References
- Top Assets
- Unused Assets
- Storage vs Usage
- Delivery Bandwidth
- Usage Analytics

Analyze by:

Asset

Asset Type

Tenant

Venue

Channel

Collection

Campaign

Date

Top Used Assets

Asset	Channels	Requests	Downloads	Status
Dubai Hero	6	284K	428	Active
Summer Promo	4	186K	214	Active
Park Logo	12	142K	1,842	Active

Underused Assets

Identify:

Assets occupying significant storage but having no usage during a configured period.

Example:

842 Assets

No usage in:

12 Months

Storage:

186 GB

Important

Unused does not mean delete.

It should create a review recommendation.

Performance Metrics

Where channel integrations provide data, DAM may display:

- impressions
- requests
- downloads
- delivery volume
- reuse count

Marketing engagement metrics such as conversions should remain owned by Marketing/CRM or the relevant analytics module and can be linked rather than duplicated.

Screen 10 — Distribution Intelligence, AI Insights & Optimization

Purpose

Provide intelligent recommendations across DAM distribution, usage, delivery, replacements, and integrations.

AI Insights

Examples:

“Six channels are still using v2.0 of Dubai Park Hero while v2.1 is the approved current version.”

“12 actively distributed assets have rights expiring within 30 days.”

“842 assets have not been referenced in 12 months and consume 186 GB.”

“Kiosk delivery latency increased 28% during the last seven days.”

“Web Hero JPEG represents 34% more bandwidth than the available WebP rendition.”

Replacement Intelligence

Example:

Asset:

Old_Dubai_Logo.png

AI identifies:

24 Active Uses

Recommended:

Dubai_Logo_2026.svg

Recommendation

Review replacement across B2C, Mobile, Kiosk and Digital Signage.

Optimization Recommendations

AI can suggest:

- Use Better Rendition
- Replace Outdated Version
- Review Unused Assets
- Review Expiring Rights
- Generate Missing Rendition
- Investigate Integration
- Optimize Delivery Format
- Predictive Indicators

Where sufficient data exists:

Delivery Failure Risk

Rights Expiry Risk

Storage Growth

Bandwidth Growth

Rendition Demand

Human Approval

Recommendations remain governed.

Board 4 — Shared Configuration Requirements

Authorized administrators should configure:

- Distribution channels
- channel types
- channel tenant/venue scope
- permitted asset types
- preferred renditions
- fallback renditions
- delivery methods
- channel eligibility rules
- CDN configuration
- caching rules
- signed URL policies
- URL expiry
- authentication requirements
- API scopes
- service accounts
- integration credentials
- integration rate limits
- usage-reference rules
- follow-current rules
- pinned-version rules
- replacement rules
- replacement approvals
- scheduled replacement
- fallback hierarchy
- failure behavior
- delivery monitoring thresholds
- API monitoring thresholds
- broken-reference rules
- alert rules
- usage analytics retention
- storage/usage thresholds
- bandwidth thresholds
- AI recommendation thresholds

Critical Architecture — Reference, Not Copy

This is the most important architecture rule for Board 4.

Other TICVAI modules should generally reference DAM assets, not copy them into their own independent storage.

Recommended architecture:

DAM Asset

DAM-IMG-008421



Asset Reference



B2C

Mobile

Ticket Media

POS

Kiosk

Marketing

If the asset changes, TICVAI can understand exactly where it is used.

If every module copies the image independently, TICVAI loses this control.

Critical Architecture — Follow Current vs Pin Version

Every usage reference should support one of two behaviors.

Follow Current

Example:

B2C homepage references:

DAM-IMG-008421 → CURRENT

Current changes:

v2.1 → v2.2

The channel can receive v2.2 after applicable governance.

Pin Version

Example:

Historical campaign:

DAM-IMG-008421 → v2.1

Even if v2.2 becomes current, the historical record remains on v2.1.

This is essential for:

- historical campaigns
- issued tickets
- financial/legal records
- archived content
- controlled brand publications

Critical Architecture — Eligibility Before Delivery

No consumer should bypass Board 3 governance.

Recommended runtime flow:

Channel Requests Asset

↓

Identify Version

↓

Board 3 Eligibility Check

↓

If eligible:

Resolve Rendition

↓

Deliver

If not eligible:

Apply Governed Fallback / Block

This means a direct API call cannot accidentally publish an expired or restricted asset.

Critical Architecture — Delivery URL vs Storage URL

These must be different.

Storage URL

Private internal location.

Never exposed.

Delivery URL

Controlled TICVAI/CDN endpoint.

Can apply:

- authorization
- expiry
- caching
- rendition resolution
- usage tracking
- revocation

Therefore:

Consumers interact with the delivery layer, not raw storage.

Critical Architecture — Replacement Propagation

When an asset must be replaced:

Old Asset



Usage Graph



Identify All Active References



Eligibility Check on Replacement



Impact Preview



Approval



Propagate



Verify Delivery



Audit

This should be a governed operation, particularly when an asset is used across many venues.

Critical Architecture — Historical Immutability

Not every asset reference should follow a replacement.

Example:

Ticket already issued on 1 September

contains Logo v1.

On 10 September, Logo v2 becomes current.

The ticket issued on 1 September should normally remain historically accurate.

Live Dynamic Content

from:

Historical Generated Output

This rule will prevent serious issues across ticketing and financial records.

Critical Architecture — Usage Graph

I recommend making the DAM Usage Graph a core service.

Example:

DAM-IMG-008421

→ B2C Homepage
 → Mobile Home
 → Kiosk Welcome
 → Ticket Template 14
 → Summer Campaign
 → Digital Signage Screen Group 8

This usage graph powers:

- replacement impact
- rights-expiry impact
- archive warnings
- version impact
- analytics
- governance
- AI recommendations

Boards 2, 3, and 4 should all consume the same graph rather than maintain separate usage records.

Integration With Ticket Media

Recommended flow:

Ticket Media Template

↓

Select DAM Asset

↓

DAM-IMG-008421

↓

Eligibility Check

↓

Required Ticket Rendition

↓

Template Rendering

For issued tickets, the resulting output/version relationship should be recorded so historical output remains traceable.

Integration With Marketing & CRM

Recommended flow:

Campaign



Select Approved DAM Collection / Asset



Channel Eligibility



Email / App / Web / Campaign



DAM Usage Registered

Marketing & CRM remains responsible for:

Audience

Campaign Performance

Conversion

while DAM remains responsible for:

Asset

Rights

Version

Delivery

Usage Reference

Integration With B2C, POS & Kiosk

These channels should request media using:

Asset ID + Usage Context

rather than hardcoded file paths.

Example:

DAM-IMG-008421

Context:

B2C_HOME_HERO

TICVAI can then resolve:

Current Eligible Version

-

Correct Rendition

-

Delivery URL

This is much more scalable than storing image URLs manually throughout the platform.

CDN Strategy

The architecture should be CDN-ready even if TICVAI initially uses a simpler storage/delivery implementation.

The delivery layer should support:

Original Storage



TICVAI Media Delivery Service



CDN



End User / Channel

Benefits include:

- faster global delivery
- reduced origin load
- caching
- optimized media
- scalable B2C traffic

The CDN provider should not become part of the DAM business logic.

API & Webhook Readiness

In addition to APIs, I recommend Board 4 architecture support future event notifications such as:

Asset Updated

New Version Current

Rights Expiring

Rights Expired

Asset Restricted

Asset Replaced

Rendition Ready

Asset Archived

Connected TICVAI modules can react without constantly polling DAM.

This can be implemented through TICVAI's wider integration/event architecture rather than a DAM-specific duplicate messaging engine.

AI Governance

AI may:

- recommend replacements
- detect outdated versions in active use
- identify delivery anomalies
- recommend optimized renditions
- predict storage growth
- predict bandwidth demand
- identify unused assets
- recommend consolidation
- identify rights-expiry exposure
- suggest fallback candidates

AI must not automatically:

- replace live assets,
- publish assets,
- override eligibility,
- change pinned versions,

- remove historical content,
- expose private assets,
- revoke business-critical delivery,
- or delete unused content.

RBAC & Governance

Permissions should distinguish:

View Distribution

View Usage Map

Configure Channel

Generate Delivery URL

Manage API Integration

View Integration Credentials

Replace Asset

Approve Replacement

Schedule Replacement

Assign Fallback

Manage CDN Settings

View Delivery Logs

View Usage Analytics

View Bandwidth/Storage Analytics

Use AI Distribution Recommendations

Export Usage Data

Tenant and venue isolation must apply to every delivery and API operation.

Board 4 — Acceptance Criteria

Board 4 is accepted when TICVAI can successfully:

Select Approved Asset → Validate Eligibility → Resolve Version → Resolve Rendition → Deliver
Securely → Register Usage → Monitor Delivery → Track Integration → Identify Impact →
Replace/Propagate → Analyze → Recommend

while maintaining the complete traceability chain:

Digital Asset → Version → Rights/Approval → Channel → Usage Reference → Rendition → Delivery →
Consumer → Performance → Replacement → Audit

The user must be able to answer, at any time:

Where is this asset currently used?

Which version is each channel using?

Which rendition is being delivered?

Is it legally and operationally eligible?

What will happen if I replace it?

Is delivery healthy?

Module 23 — Complete Architecture

With Board 4, the proposed Digital Asset Management module is complete:

Board 1 — Digital Asset Library & Organization

10 screens

Board 2 — AI Asset Intelligence, Versioning & Rendition Management

10 screens

Board 3 — Asset Governance, Rights, Approval & Secure Sharing

10 screens

Board 4 — Asset Distribution, Delivery, Integration & Analytics

10 screens

Total: 4 Boards × 10 Screens = 40 Screens

The complete DAM lifecycle is therefore:

**Upload → Organize → Understand → Version → Transform → Govern → Approve → Distribute → Track →
Replace → Analyze**

This gives development team a single DAM architecture instead of separate image libraries being built independently inside Ticketing, Marketing, B2C, POS, Kiosk, and other TICVAI modules.